Proteins are also involved in highly ordered metabolic pathways, and a defect "upstream" can mask or prevent expression of other alleles "downstream." This condition is known as **epistasis** ("standing upon"), and the upstream gene is said to be epistatic to the downstream one. For instance, on blood cells, the well-known ABO markers are actually branched sugars attached to proteins embedded in the surface of the cell. In order for the cell to attach these sugars, it must first express a gene (called fucosyltransferase) that attaches one sugar group (fucose) that is common to all blood types. Absence of functional fucosyltransferase prevents the expression of the ABO alleles. SEE ALSO COLOR VISION; CROSSING OVER; DISEASE, GENETICS OF; EPISTASIS; FERTILIZATION; GROWTH DISORDERS; HARDY-WEINBERG EQUILIBRIUM; HEMOPHILIA; IMPRINTING; MEIOSIS; MENDELIAN GENETICS; MITOCHONDRIAL DISEASES; MOSAICISM; MUSCULAR DYSTROPHY; PEDIGREE; PLEIOTROPY; TAY-SACHS DISEASE; TRIPLET REPEAT DISEASE.

epistasis supression of a characteristic of one gene by the action of another gene

Richard Robinson

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Intelligence

The roles of genes and environment in the determination of intelligence have been controversial for more than 100 years. Studies of the question have often been marred by untested assumptions, poor design, and even racism, faults that more modern studies have striven to avoid. Nonetheless, examining the biology of intelligence is an enterprise that continues to be fraught with difficulty, and there remains no real consensus even on how to define the term.

IQ Tests

Conventional measures of intelligence are obtained using standard tests, called intelligence quotient tests or, more commonly, IQ tests. These tests have been shown to be reliable and valid. Reliability means that they measure the same thing from person to person, whereas validity means that they measure what they claim to measure. IQ tests measure a person's ability to reason and to solve problems. These abilities are frequently called general cognitive ability, or "g."

Almost all genetic studies of the **heritability** of intelligence (how much is due to genetics and how much is due to the environment) have been obtained from IQ tests. To understand the studies, therefore, it is important to understand what IQ tests measure, and how their use and interpretation have changed over time.

heritability proportion of variability due to genes; ability to be inherited

The standard IQ-measurement approach to intelligence is among the oldest of approaches and probably began in 1876, when Francis Galton investigated how much the similarity between twins changed as they developed

MENSA is an international organization whose only membership criteria is a score in the top 2 percent of the population on a standard IQ test.

over time. Galton's study was concerned with measuring psychophysical abilities, such as strength of handgrip or visual acuity. The concept of general cognitive ability was first described by Charles Spearman in 1904. Later, Alfred Binet and Theophile Simon (1916) evaluated intelligence based on judgment, involving adaptation to the environment, direction of one's efforts, and self-criticism.

Most standard test results now include three scores: VIQ, PIQ, and FSIQ. The VIQ score measures verbal ability (verbal IQ), PIQ measures performance ability (performance IQ), and FSIQ provides an overall measurement (full scale IQ). Commonly used IQ tests include the Stanford-Binet Intelligence Scale, the Wechsler Intelligence Scale for Children (WISC), and the Wechsler Adult Intelligence Scales. The results achieved by individual testtakers on one of these IQ tests are likely to be similar to the results they achieve on the others, and they all aim to measure general cognitive ability (among other things). Measures of scholastic achievement, such as the SAT and the ACT correlate highly with "g."

Environmental Effects on Intelligence

The study of intelligence must take environmental effects into account. The Flynn effect describes a phenomenon that indicates that IQ has increased about 3 points per decade over the last fifty years, with children scoring higher than parents in each generation. This increase has been linked to multiple environmental factors, including better nutrition, increased schooling, higher educational attainment of parents, less childhood disease, more complex environmental stimulation, lower birth rates, and a variety of other factors.

Males and females have equivalent "g" scores. The question of racial differences and IQ arose when a 10 point IQ difference between African Americans and Americans of European descent was documented. Two adoption studies indicate that the effect may be in part related to environment factors, including culture. Also, environmental differences similar to those identified with the Flynn effect can be postulated. Studies of black Caribbean children and English children raised in an orphanage in England found that the black Caribbean children had higher IQs than the English children, with mixed racial children in between. A study comparing black children adopted by white families and those who were adopted into black families in the United States showed that black children raised by whites had higher IQ scores, again suggesting that the environment played a role.

Expanded Concepts of Intelligence

Many of the standard measures of IQ, such as the WISC and the Stanford-Binet, have changed their content over the years. Although they both still report verbal, performance, and total scores, the Wechsler model now offers scores for four additional factors (verbal comprehension, perceptual organization, processing speed, and freedom from distractibility). The Stanford-Binet also yields additional scores, including abstract-visual reasoning, quantitative reasoning, and short-term memory.

However, the majority of research into genetic and environmental variance in IQ has centered on the assumption that general cognitive ability is the essence of intelligence. Newer tests that measure specific abilities have not been included in genetic studies. These include, for example, tests that measure creativity in a model for intelligence. The addition of new factors in the Wechsler and Stanford-Binet IQ tests represents a trend toward a broader approach to IQ, and away from the notion that IQ can be understood by the single factor, "g."

Family, Twin, and Adoption Studies

Genetic studies have traditionally used models that evaluate how much of the variability in IQ is due to genes and how much is associated with environment. These studies include family studies, twin studies, and adoption studies.

General cognitive ability runs in families. For first-degree relatives (parents, children, brothers, sisters) living together, correlations of "g" for over 8,000 parent-offspring pairs averaged 0.43 (0.0 is no correlation, 1.0 is complete correlation). For more than 25,000 sibling pairs, "g" correlations averaged 0.47. Heritability estimates range from 40 to 80 percent, meaning that 40 to 80 percent of "g" is due to genes.

In twin studies of over 10,000 pairs of twins, monozygotic (genetically identical) twins averaged an 0.85 correlation of "g," whereas for dizygotic (fraternal, like brothers or sisters) same-sex twins the "g" correlations were 0.60. These twin studies suggest that the heritability (genetic effect) accounts for about half of the variance in "g" scores.

Adoption studies also provide evidence for substantial heritability of "g." The "g" estimate for identical twins raised apart is similar to that of identical twins raised together, proving that for genetically identical individuals, environmental differences did not affect "g." The Colorado Adoption Study (CAP) of first-degree relatives who were adopted also indicated significant heritability of "g." Thus, classical genetic studies indicate that there is a statistically significant and substantial genetic influence on "g."

Newer genetic research on general cognitive ability has focused on developmental changes in IQ, multivariate relations (contributions of multiple factors) among cognitive abilities, and specific genes responsible for the heritability of "g." Developmental changes over time were first studied by Galton in 1876. The CAP study was conducted over twenty-five years and evaluated 245 children who had been separated from their parents at birth and adopted by one month of age. This study, and others, showed that the variance in "g" due to environment for an adopted child in his or her adoptive family is largely unconnected with the shared adoptive family upbringing, that is, a shared parent-sibling environment. For adoptive parents and their adopted children, the parent-offspring correlations for heritability were around zero. For adopted children and their biologic mothers or for children raised with their biologic parents, heritability was the same, increasing with age.

Recent studies indicate that heritability increases over time, with infant measures of about 20 percent, childhood measures at 40 percent, and adult measures reaching 60 percent. Why is there an age effect for the heritability of "g"? Part of this could be due to different genes being expressed over time, as the brain develops. The stability of the heritability measure correlates with changes in brain development, with "maturity" of brain structure achieved after adolescence. Also, it is likely that small gene effects early in life become larger as children and adolescents select or create environments that foster their strengths.

Multivariate relations among cognitive abilities affect more than general cognitive ability as measured by "g." Current models of cognitive abilities include specific components such as spatial and verbal abilities, speed of processing, and memory abilities. Less is known about the heritabilities of these specific cognitive skills. They also show substantial genetic influence, although this influence is less than what has been found for "g." Multivariate genetic analyses indicate that the same genetic factors influence different abilities. In other words, a specific gene found to be associated with verbal ability may also be associated with spatial ability and other specific cognitive abilities. Four studies have shown that genetic effects on measures of school achievement are highly correlated with genetic effects on "g." Also, discrepancies between school achievement and "g," as occurs with underachievers, are predominantly of environmental origin.

Genes for Intelligence

The search for specific genes associated with IQ is proceeding at a rapid pace with the completion of the Human Genome Project. While defects in single genes, such as the fragile X gene, can cause mental retardation, the heritability of general cognitive ability is most likely due to multiple genes of small effect (called quantitative trait loci, or QTLs) rather than a single gene of large effect. QTLs contribute additively and interchangeably to intelligence.

Genetic studies have identified QTLs associated with "g" on chromosomes 4 and 6. These studies involved both children with high "g" and children with average "g." QTLs on chromosome 6 have been identified and shown to be active in the regions of the brain involved in learning and memory. The gene identified is for insulin-like growth factor 2 receptor, or IGF2R, the exact function of which is still unknown. One allele (alternative form) of IGR2R was found to be present 30 percent of the time in two groups of children with high "g." This was twice the frequency of its occurrence in two groups of children with average "g," and these findings have been successfully replicated in other studies. QTLs associated with "g" have also been identified on chromosome 4. Future identification of QTLs will allow geneticists to begin to answer questions about IQ and development and gene-environment interaction directly, rather than relying on less specific family, adoption, and twin studies.

In summary, intelligence measurements ranging from specific cognitive abilities to "g" have a complex relationship. Genetic contributions are large, and heritability increases with age. Heritability remains high for verbal abilities during adulthood. Finally, the identification of QTLs associated with "g" and with specific cognitive abilities is just beginning. SEE ALSO Behavior; Complex Traits; Eugenics; Fragile X syndrome; Genetic Discrimination; Quantitative Traits; Twins.

Harry Wright and Ruth Abramson

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Internet

Biologists often use two terms to describe alternative approaches for conducting experiments. "In vitro" (Latin for "in glass") refers to experiments typically carried out in test tubes with purified biochemicals. "In vivo" ("in life") experiments are performed directly on living organisms. In recent years, the indispensable use of computers and the Internet for genetic and molecular biology research has introduced a new term into the language: "in silico" ("in silicon"), referring to the silicon used to manufacture computer chips. In silico genetics experiments are those that are performed with a computer, often involving analysis of DNA or protein sequences over the Internet.

Geneticists and molecular biologists use the Internet much the same way most people do, communicating data and results through e-mail and discussion groups and sharing information on Web sites, for instance. They also make wide use of powerful Internet-based databases and analytical tools. Researchers are determining the DNA sequences of entire **genomes** at an ever accelerating pace, and are devising methods for cataloging entire sets of proteins (termed "proteomes") expressed in organisms. The databases to store all this information are growing at an equal pace, and the computer tools to sort through all the data are becoming increasingly sophisticated.

genomes the total genetic material in cells or organisms

One of the most important Web sites for biological computer analysis (sometimes called **bioinformatics**) is that of the National Center for Biotechnology Information (NCBI), a part of the National Library of Medicine, which, in turn, is part of the National Institutes of Health. The NCBI Web site hosts DNA and protein sequence databases, protein three-dimensional structure databases, scientific literature databases, and search engines for retrieving files of interest. All of these resources are freely accessible to anyone on the Internet.

bioinformatics use of information technology to analyze biological data

Of all the powerful analytical tools available at NCBI, probably the most important and heavily used is a set of computer programs called BLAST, for Basic Local Alignment Search Tool. BLAST can rapidly search many sequence databases to see whether any DNA or protein sequence (a "query sequence," supplied by the user) is similar to other sequences. Since sequence similarity usually suggests that two proteins or DNA molecules are homologous (i.e., that they are evolutionarily related and therefore may have—or encode proteins—with similar functions), discovering a blast match between an unknown protein or nucleic acid sequence and a well-characterized sequence provides an immediate clue about the function of the unknown sequence. An important scientific discovery that, in the past, may have taken many years of in vitro and in vivo analysis to arrive at is now made in a few seconds, with this simple in silico experiment. SEE ALSO Bioinformatics; Genome; Genomics; Homology; Proteomics; Sequencing DNA.

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Glossary

α the Greek letter alpha

β the Greek letter beta

γ the Greek letter gamma

λ the Greek letter lambda

σ the Greek letter sigma

E. coli the bacterium *Escherichia coli*

"-ase" suffix indicating an enzyme

acidic having the properties of an acid; the opposite of basic

acrosomal cap tip of sperm cell that contains digestive enzymes for penetrating the egg

adenoma a tumor (cell mass) of gland cells

aerobic with oxygen, or requiring it

agar gel derived from algae

agglutinate clump together

aggregate stick together

algorithm procedure or set of steps

allele a particular form of a gene

allelic variation presence of different gene forms (alleles) in a population

allergen substance that triggers an allergic reaction

allolactose "other lactose"; a modified form of lactose

amino acid a building block of protein

amino termini the ends of a protein chain with a free NH_2 group

amniocentesis removal of fluid from the amniotic sac surrounding a fetus, for diagnosis

amplify produce many copies of, multiply

anabolic steroids hormones used to build muscle mass

anaerobic without oxygen or not requiring oxygen

androgen testosterone or other masculinizing hormone

anemia lack of oxygen-carrying capacity in the blood

aneuploidy abnormal chromosome numbers

angiogenesis growth of new blood vessels

anion negatively charged ion

anneal join together

anode positive pole

anterior front

antibody immune-system protein that binds to foreign molecules

antidiuretic a substance that prevents water loss

antigen a foreign substance that provokes an immune response

antigenicity ability to provoke an immune response

apoptosis programmed cell death

Archaea one of three domains of life, a type of cell without a nucleus

archaeans members of one of three domains of life, have types of cells without a nucleus

aspirated removed with a needle and syringe

aspiration inhalation of fluid or solids into the lungs

association analysis estimation of the relationship between alleles or genotypes and disease

asymptomatic without symptoms

ATP adenosine triphosphate, a high-energy compound used to power cell processes

ATPase an enzyme that breaks down ATP, releasing energy

attenuation weaken or dilute

atypical irregular

autoimmune reaction of the immune system to the body's own tissues

autoimmunity immune reaction to the body's own tissues

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

autosome a chromosome that is not sex-determining (not X or Y)

axon the long extension of a nerve cell down which information flows

bacteriophage virus that infects bacteria

basal lowest level

base pair two nucleotides (either DNA or RNA) linked by weak bonds

basic having the properties of a base; opposite of acidic

benign type of tumor that does not invade surrounding tissue

binding protein protein that binds to another molecule, usually either DNA or protein

biodiversity degree of variety of life

bioinformatics use of information technology to analyze biological data

biolistic firing a microscopic pellel into a biological sample (from biological/ballistic)

biopolymers biological molecules formed from similar smaller molecules, such as DNA or protein

biopsy removal of tissue sample for diagnosis

biotechnology production of useful products

bipolar disorder psychiatric disease characterized by alternating mania and depression

blastocyst early stage of embryonic development

brackish a mix of salt water and fresh water

breeding analysis analysis of the offspring ratios in breeding experiments

buffers substances that counteract rapid or wide pH changes in a solution

Cajal Ramon y Cajal, Spanish neuroanatomist

carcinogens substances that cause cancer

carrier a person with one copy of a gene for a recessive trait, who therefore does not express the trait

catalyst substance that speeds a reaction without being consumed (e.g., enzyme)

catalytic describes a substance that speeds a reaction without being consumed

catalyze aid in the reaction of

cathode negative pole

cDNA complementary DNA

cell cycle sequence of growth, replication and division that produces new cells

centenarian person who lives to age 100

centromere the region of the chromosome linking chromatids

cerebrovascular related to the blood vessels in the brain

cerebrovascular disease stroke, aneurysm, or other circulatory disorder affecting the brain

charge density ratio of net charge on the protein to its molecular mass

chemotaxis movement of a cell stimulated by a chemical attractant or repellent

chemotherapeutic use of chemicals to kill cancer cells

chloroplast the photosynthetic organelle of plants and algae

chondrocyte a cell that forms cartilage

chromatid a replicated chromosome before separation from its copy

chromatin complex of DNA, histones, and other proteins, making up chromosomes

ciliated protozoa single-celled organism possessing cilia, short hair-like extensions of the cell membrane

circadian relating to day or day length

cleavage hydrolysis

cleave split

clinical trials tests performed on human subjects

codon a sequence of three mRNA nucleotides coding for one amino acid

Cold War prolonged U.S.-Soviet rivalry following World War II

colectomy colon removal

colon crypts part of the large intestine

complementary matching opposite, like hand and glove

conformation three-dimensional shape

congenital from birth

conjugation a type of DNA exchange between bacteria

cryo-electron microscope electron microscope that integrates multiple images to form a three-dimensional model of the sample

cryopreservation use of very cold temperatures to preserve a sample

cultivars plant varieties resulting from selective breeding

cytochemist chemist specializing in cellular chemistry

cytochemistry cellular chemistry

cytogenetics study of chromosome structure and behavior

cytologist a scientist who studies cells

cytokine immune system signaling molecule

cytokinesis division of the cell's cytoplasm

cytology the study of cells

cytoplasm the material in a cell, excluding the nucleus

cytosol fluid portion of a cell, not including the organelles

de novo entirely new

deleterious harmful

dementia neurological illness characterized by impaired thought or awareness

demography aspects of population structure, including size, age distribution, growth, and other factors

denature destroy the structure of

deoxynucleotide building block of DNA

dimerize linkage of two subunits

dimorphism two forms

diploid possessing pairs of chromosomes, one member of each pair derived from each parent

disaccharide two sugar molecules linked together

dizygotic fraternal or nonidentical

DNA deoxyribonucleic acid

domains regions

dominant controlling the phenotype when one allele is present

dopamine brain signaling chemical

dosage compensation equalizing of expression level of X-chromosome genes between males and females, by silencing one X chromosome in females or amplifying expression in males

ecosystem an ecological community and its environment

ectopic expression expression of a gene in the wrong cells or tissues

electrical gradient chemiosmotic gradient

electrophoresis technique for separation of molecules based on size and charge

eluting exiting

embryogenesis development of the embryo from a fertilized egg

endangered in danger of extinction throughout all or a significant portion of a species' range

endogenous derived from inside the organism

endometriosis disorder of the endometrium, the lining of the uterus

endometrium uterine lining

endonuclease enzyme that cuts DNA or RNA within the chain

endoplasmic reticulum network of membranes within the cell

endoscope tool used to see within the body

endoscopic describes procedure wherein a tool is used to see within the body

endosymbiosis symbiosis in which one partner lives within the other

enzyme a protein that controls a reaction in a cell

epidemiologic the spread of diseases in a population

epidemiologists people who study the incidence and spread of diseases in a population

epidemiology study of incidence and spread of diseases in a population

epididymis tube above the testes for storage and maturation of sperm

epigenetic not involving DNA sequence change

epistasis suppression of a characteristic of one gene by the action of another gene

epithelial cells one of four tissue types found in the body, characterized by thin sheets and usually serving a protective or secretory function

Escherichia coli common bacterium of the human gut, used in research as a model organism

estrogen female horomone

et al. "and others"

ethicists a person who writes and speaks about ethical issues

etiology causation of disease, or the study of causation

eubacteria one of three domains of life, comprising most groups previously classified as bacteria

eugenics movement to "improve" the gene pool by selective breeding

eukaryote organism with cells possessing a nucleus

eukaryotic describing an organism that has cells containing nuclei

ex vivo outside a living organism

excise remove; cut out

excision removal

exogenous from outside

exon coding region of genes

exonuclease enzyme that cuts DNA or RNA at the end of a strand

expression analysis whole-cell analysis of gene expression (use of a gene to create its RNA or protein product)

fallopian tubes tubes through which eggs pass to the uterus

fermentation biochemical process of sugar breakdown without oxygen

fibroblast undifferentiated cell normally giving rise to connective tissue cells

fluorophore fluorescent molecule

forensic related to legal proceedings

founder population

fractionated purified by separation based on chemical or physical properties

fraternal twins dizygotic twins who share 50 percent of their genetic material

frontal lobe one part of the forward section of the brain, responsible for planning, abstraction, and aspects of personality

gamete reproductive cell, such as sperm or egg

gastrulation embryonic stage at which primitive gut is formed

gel electrophoresis technique for separation of molecules based on size and charge

gene expression use of a gene to create the corresponding protein

genetic code the relationship between RNA nucleotide triplets and the amino acids they cause to be added to a growing protein chain

genetic drift evolutionary mechanism, involving random change in gene frequencies

genetic predisposition increased risk of developing diseases

genome the total genetic material in a cell or organism

genomics the study of gene sequences

genotype set of genes present

geothermal related to heat sources within Earth

germ cell cell creating eggs or sperm

germ-line cells giving rise to eggs or sperm

gigabase one billion bases (of DNA)

glucose sugar

glycolipid molecule composed of sugar and fatty acid

glycolysis the breakdown of the six-carbon carbohydrates glucose and fructose

glycoprotein protein to which sugars are attached

Golgi network system in the cell for modifying, sorting, and delivering proteins

gonads testes or ovaries

gradient a difference in concentration between two regions

Gram negative bacteria bacteria that do not take up Gram stain, due to membrane structure

Gram positive able to take up Gram stain, used to classify bacteria

gynecomastia excessive breast development in males

haploid possessing only one copy of each chromosome

haplotype set of alleles or markers on a short chromosome segment

hematopoiesis formation of the blood

hematopoietic blood-forming

heme iron-containing nitrogenous compound found in hemoglobin

hemolysis breakdown of the blood cells

hemolytic anemia blood disorder characterized by destruction of red blood cells

hemophiliacs a person with hemophilia, a disorder of blood clotting

herbivore plant eater

heritability proportion of variability due to genes; ability to be inherited

heritability estimates how much of what is observed is due to genetic factors

heritable genetic

heterochromatin condensed portion of chromosomes

heterozygote an individual whose genetic information contains two different forms (alleles) of a particular gene

heterozygous characterized by possession of two different forms (alleles) of a particular gene

high-throughput rapid, with the capacity to analyze many samples in a short time

histological related to tissues

histology study of tissues

histone protein around which DNA winds in the chromosome

homeostasis maintenance of steady state within a living organism

homologous carrying similar genes

homologues chromosomes with corresponding genes that pair and exchange segments in meiosis

homozygote an individual whose genetic information contains two identical copies of a particular gene

homozygous containing two identical copies of a particular gene

hormones molecules released by one cell to influence another

hybrid combination of two different types

hybridization (molecular) base-pairing among DNAs or RNAs of different origins

hybridize to combine two different species

hydrogen bond weak bond between the H of one molecule or group and a nitrogen or oxygen of another

hydrolysis splitting with water

hydrophilic "water-loving"

hydrophobic "water hating," such as oils

hydrophobic interaction attraction between portions of a molecule (especially a protein) based on mutual repulsion of water

hydroxyl group chemical group consisting of -OH

hyperplastic cell cell that is growing at an increased rate compared to normal cells, but is not yet cancerous

hypogonadism underdeveloped testes or ovaries

hypothalamus brain region that coordinates hormone and nervous systems

hypothesis testable statement

identical twins monozygotic twins who share 100 percent of their genetic material

immunogenicity likelihood of triggering an immune system defense

immunosuppression suppression of immune system function

immunosuppressive describes an agent able to suppress immune system function

in vitro "in glass"; in lab apparatus, rather than within a living organism

in vivo "in life"; in a living organism, rather than in a laboratory apparatus

incubating heating to optimal temperature for growth

informed consent knowledge of risks involved

insecticide substance that kills insects

interphase the time period between cell divisions

intra-strand within a strand

intravenous into a vein

intron untranslated portion of a gene that interrupts coding regions

karyotype the set of chromosomes in a cell, or a standard picture of the chromosomes

kilobases units of measure of the length of a nucleicacid chain; one kilobase is equal to 1,000 base pairs

kilodalton a unit of molecular weight, equal to the weight of 1000 hydrogen atoms

kinase an enzyme that adds a phosphate group to another molecule, usually a protein

knocking out deleting of a gene or obstructing gene expression

laparoscope surgical instrument that is inserted through a very small incision, usually guided by some type of imaging technique

latent present or potential, but not apparent

lesion damage

ligand a molecule that binds to a receptor or other molecule

ligase enzyme that repairs breaks in DNA

ligate join together

linkage analysis examination of co-inheritance of disease and DNA markers, used to locate disease genes

lipid fat or wax-like molecule, insoluble in water

loci/locus site(s) on a chromosome

longitudinally lengthwise

lumen the space within the tubes of the endoplasmic reticulum

lymphocytes white blood cells

lyse break apart

lysis breakage

macromolecular describes a large molecule, one composed of many similar parts

macromolecule large molecule such as a protein, a carbohydrate, or a nucleic acid

macrophage immune system cell that consumes foreign material and cellular debris

malignancy cancerous tissue

malignant cancerous; invasive tumor

media (bacteria) nutrient source

meiosis cell division that forms eggs or sperm

melanocytes pigmented cells

meta-analysis analysis of combined results from multiple clinical trials

metabolism chemical reactions within a cell

metabolite molecule involved in a metabolic pathway

metaphase stage in mitosis at which chromosomes are aligned along the cell equator

metastasis breaking away of cancerous cells from the initial tumor

metastatic cancerous cells broken away from the initial tumor

methylate add a methyl group to

methylated a methyl group, CH_3, added

methylation addition of a methyl group, CH_3

microcephaly reduced head size

microliters one thousandth of a milliliter

micrometer 1/1000 meter

microsatellites small repetitive DNA elements dispersed throughout the genome

microtubule protein strands within the cell, part of the cytoskeleton

miscegenation racial mixing

mitochondria energy-producing cell organelle

mitogen a substance that stimulates mitosis

mitosis separation of replicated chromosomes

molecular hybridization base-pairing among DNAs or RNAs of different origins

molecular systematics the analysis of DNA and other molecules to determine evolutionary relationships

monoclonal antibodies immune system proteins derived from a single B cell

monomer "single part"; monomers are joined to form a polymer

monosomy gamete that is missing a chromosome

monozygotic genetically identical

morphologically related to shape and form

morphology related to shape and form

mRNA messenger RNA

mucoid having the properties of mucous

mucosa outer covering designed to secrete mucus, often found lining cavities and internal surfaces

mucous membranes nasal passages, gut lining, and other moist surfaces lining the body

multimer composed of many similar parts

multinucleate having many nuclei within a single cell membrane

mutagen any substance or agent capable of causing a change in the structure of DNA

mutagenesis creation of mutations

mutation change in DNA sequence

nanometer 10^{-9}(exp) meters; one billionth of a meter

nascent early-stage

necrosis cell death from injury or disease

nematode worm of the Nematoda phylum, many of which are parasitic

neonatal newborn

neoplasms new growths

neuroimaging techniques for making images of the brain

neurological related to brain function or disease

neuron nerve cell

neurotransmitter molecule released by one neuron to stimulate or inhibit a neuron or other cell

non-polar without charge separation; not soluble in water

normal distribution distribution of data that graphs as a bell-shaped curve

Northern blot a technique for separating RNA molecules by electrophoresis and then identifying a target fragment with a DNA probe

Northern blotting separating RNA molecules by electrophoresis and then identifying a target fragment with a DNA probe

nuclear DNA DNA contained in the cell nucleus on one of the 46 human chromosomes; distinct from DNA in the mitochondria

nuclear membrane membrane surrounding the nucleus

nuclease enzyme that cuts DNA or RNA

nucleic acid DNA or RNA

nucleoid region of the bacterial cell in which DNA is located

nucleolus portion of the nucleus in which ribosomes are made

nucleoplasm material in the nucleus

nucleoside building block of DNA or RNA, composed of a base and a sugar

nucleoside triphosphate building block of DNA or RNA, composed of a base and a sugar linked to three phosphates

nucleosome chromosome structural unit, consisting of DNA wrapped around histone proteins

nucleotide a building block of RNA or DNA

ocular related to the eye

oncogene gene that causes cancer

oncogenesis the formation of cancerous tumors

oocyte egg cell

open reading frame DNA sequence that can be translated into mRNA; from start sequence to stop sequence

opiate opium, morphine, and related compounds

organelle membrane-bound cell compartment

organic composed of carbon, or derived from living organisms; also, a type of agriculture stressing soil fertility and avoidance of synthetic pesticides and fertilizers

osmotic related to differences in concentrations of dissolved substances across a permeable membrane

ossification bone formation

osteoarthritis a degenerative disease causing inflammation of the joints

osteoporosis thinning of the bone structure

outcrossing fertilizing between two different plants

oviduct a tube that carries the eggs

ovulation release of eggs from the ovaries

ovules eggs

ovum egg

oxidation chemical process involving reaction with oxygen, or loss of electrons

oxidized reacted with oxygen

pandemic disease spread throughout an entire population

parasites organisms that live in, with, or on another organism

pathogen disease-causing organism

pathogenesis pathway leading to disease

pathogenic disease-causing

pathogenicity ability to cause disease

pathological altered or changed by disease

pathology disease process

pathophysiology disease process

patient advocate a person who safeguards patient rights or advances patient interests

PCR polymerase chain reaction, used to amplify DNA

pedigrees sets of related individuals, or the graphic representation of their relationships

peptide amino acid chain

peptide bond bond between two amino acids

percutaneous through the skin

phagocytic cell-eating

phenotype observable characteristics of an organism

phenotypic related to the observable characteristics of an organism

pheromone molecule released by one organism to influence another organism's behavior

phosphate group PO_4 group, whose presence or absence often regulates protein action

phosphodiester bond the link between two nucleotides in DNA or RNA

phosphorylating addition of phosphate group (PO_4)

phosphorylation addition of the phosphate group PO_4^{3-}

phylogenetic related to the evolutionary development of a species

phylogeneticists scientists who study the evolutionary development of a species

phylogeny the evolutionary development of a species

plasma membrane outer membrane of the cell

plasmid a small ring of DNA found in many bacteria

plastid plant cell organelle, including the chloroplast

pleiotropy genetic phenomenon in which alteration of one gene leads to many phenotypic effects

point mutation gain, loss, or change of one to several nucleotides in DNA

polar partially charged, and usually soluble in water

pollen male plant sexual organ

polymer molecule composed of many similar parts

polymerase enzyme complex that synthesizes DNA or RNA from individual nucleotides

polymerization linking together of similar parts to form a polymer

polymerize to link together similar parts to form a polymer

polymers molecules composed of many similar parts

polymorphic occurring in several forms

polymorphism DNA sequence variant

polypeptide chain of amino acids

polyploidy presence of multiple copies of the normal chromosome set

population studies collection and analysis of data from large numbers of people in a population, possibly including related individuals

positional cloning the use of polymorphic genetic markers ever closer to the unknown gene to track its inheritance in CF families

posterior rear

prebiotic before the origin of life

precursor a substance from which another is made

prevalence frequency of a disease or condition in a population

primary sequence the sequence of amino acids in a protein; also called primary structure

primate the animal order including humans, apes, and monkeys

primer short nucleotide sequence that helps begin DNA replication

primordial soup hypothesized prebiotic environment rich in life's building blocks

probe molecule used to locate another molecule

procarcinogen substance that can be converted into a carcinogen, or cancer-causing substance

procreation reproduction

progeny offspring

prokaryote a single-celled organism without a nucleus

promoter DNA sequence to which RNA polymerase binds to begin transcription

promutagen substance that, when altered, can cause mutations

pronuclei egg and sperm nuclei before they fuse during fertilization

proprietary exclusively owned; private

proteomic derived from the study of the full range of proteins expressed by a living cell

proteomics the study of the full range of proteins expressed by a living cell

protists single-celled organisms with cell nuclei

protocol laboratory procedure

protonated possessing excess H^+ ions; acidic

pyrophosphate free phosphate group in solution

quiescent non-dividing

radiation high energy particles or waves capable of damaging DNA, including X rays and gamma rays

recessive requiring the presence of two alleles to control the phenotype

recombinant DNA DNA formed by combining segments of DNA, usually from different types of organisms

recombining exchanging genetic material

replication duplication of DNA

restriction enzyme an enzyme that cuts DNA at a particular sequence

retina light-sensitive layer at the rear of the eye

retroviruses RNA-containing viruses whose genomes are copied into DNA by the enzyme reverse transcriptase

reverse transcriptase enzyme that copies RNA into DNA

ribonuclease enzyme that cuts RNA

ribosome protein-RNA complex at which protein synthesis occurs

ribozyme RNA-based catalyst

RNA ribonucleic acid

RNA polymerase enzyme complex that creates RNA from DNA template

RNA triplets sets of three nucleotides

salinity of, or relating to, salt

sarcoma a type of malignant (cancerous) tumor

scanning electron microscope microscope that produces images with depth by bouncing electrons off the surface of the sample

sclerae the "whites" of the eye

scrapie prion disease of sheep and goats

segregation analysis statistical test to determine pattern of inheritance for a trait

senescence a state in a cell in which it will not divide again, even in the presence of growth factors

senile plaques disease

serum (pl. sera) fluid portion of the blood

sexual orientation attraction to one sex or the other

somatic nonreproductive; not an egg or sperm

Southern blot a technique for separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe

Southern blotting separating DNA fragments by electrophoresis and then identifying a target fragment with a DNA probe

speciation the creation of new species

spindle football-shaped structure that separates chromosomes in mitosis

spindle fiber protein chains that separate chromosomes during mitosis

spliceosome RNA-protein complex that removes introns from RNA transcripts

spontaneous non-inherited

sporadic caused by new mutations

stem cell cell capable of differentiating into multiple other cell types

stigma female plant sexual organ

stop codon RNA triplet that halts protein synthesis

striatum part of the midbrain

subcutaneous under the skin

sugar glucose

supercoiling coiling of the helix

symbiont organism that has a close relationship (symbiosis) with another

symbiosis a close relationship between two species in which at least one benefits

symbiotic describes a close relationship between two species in which at least one benefits

synthesis creation

taxon/taxa level(s) of classification, such as kingdom or phylum

taxonomical derived from the science that identifies and classifies plants and animals

taxonomist a scientist who identifies and classifies organisms

telomere chromosome tip

template a master copy

tenets generally accepted beliefs

terabyte a trillion bytes of data

teratogenic causing birth defects

teratogens substances that cause birth defects

thermodynamics process of energy transfers during reactions, or the study of these processes

threatened likely to become an endangered species

topological describes spatial relations, or the study of these relations

topology spatial relations, or the study of these relations

toxicological related to poisons and their effects

transcript RNA copy of a gene

transcription messenger RNA formation from a DNA sequence

transcription factor protein that increases the rate of transcription of a gene

transduction conversion of a signal of one type into another type

transgene gene introduced into an organism

transgenics transfer of genes from one organism into another

translation synthesis of protein using mRNA code

translocation movement of chromosome segment from one chromosome to another

transposable genetic element DNA sequence that can be copied and moved in the genome

transposon genetic element that moves within the genome

trilaminar three-layer

triploid possessing three sets of chromosomes

trisomics mutants with one extra chromosome

trisomy presence of three, instead of two, copies of a particular chromosome

tumor mass of undifferentiated cells; may become cancerous

tumor suppressor genes cell growths

tumors masses of undifferentiated cells; may become cancerous

vaccine protective antibodies

vacuole cell structure used for storage or related functions

van der Waal's forces weak attraction between two different molecules

vector carrier

vesicle membrane-bound sac

virion virus particle

wet lab laboratory devoted to experiments using solutions, cell cultures, and other "wet" substances

wild-type most common form of a trait in a population

Wilm's tumor a cancerous cell mass of the kidney

X ray crystallography use of X rays to determine the structure of a molecule

xenobiotic foreign biological molecule, especially a harmful one

zygote fertilized egg

Topic Outline

APPLICATIONS TO OTHER FIELDS

Agricultural Biotechnology
Biopesticides
Bioremediation
Biotechnology
Conservation Biology: Genetic Approaches
DNA Profiling
Genetically Modified Foods
Molecular Anthropology
Pharmacogenetics and Pharmacogenomics
Plant Genetic Engineer
Public Health, Genetic Techniques in
Transgenic Animals
Transgenic Microorganisms
Transgenic Plants

BACTERIAL GENETICS

Escherichia coli (*E. coli* bacterium)
Ames Test
Antibiotic Resistance
Chromosome, Prokaryotic
Cloning Genes
Conjugation
Eubacteria
Microbiologist
Overlapping Genes
Plasmid
Transduction
Transformation
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Viroids and Virusoids
Virus

BASIC CONCEPTS

Biotechnology
Crossing Over
Disease, Genetics of
DNA
DNA Structure and Function, History
Fertilization
Gene
Genetic Code
Genetics
Genome
Genotype and Phenotype
Homology
Human Genome Project
Inheritance Patterns
Meiosis
Mendelian Genetics
Mitosis
Mutation
Nucleotide
Plasmid
Population Genetics
Proteins
Recombinant DNA
Replication
RNA
Transcription
Translation

BIOTECHNOLOGY

Agricultural Biotechnology
Biopesticides
Bioremediation
Biotechnology
Biotechnology and Genetic Engineering, History
Biotechnology: Ethical Issues
Cloning Genes
Cloning Organisms
DNA Vaccines

Genetically Modified Foods
HPLC: High-Performance Liquid Chromatography
Pharmaceutical Scientist
Plant Genetic Engineer
Polymerase Chain Reaction
Recombinant DNA
Restriction Enzymes
Reverse Transcriptase
Transgenic Animals
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Transgenic Plants

CAREERS

Attorney
Bioinformatics
Clinical Geneticist
College Professor
Computational Biologist
Conservation Geneticist
Educator
Epidemiologist
Genetic Counselor
Geneticist
Genomics Industry
Information Systems Manager
Laboratory Technician
Microbiologist
Molecular Biologist
Pharmaceutical Scientist
Physician Scientist
Plant Genetic Engineer
Science Writer
Statistical Geneticist
Technical Writer

CELL CYCLE

Apoptosis
Balanced Polymorphism
Cell Cycle
Cell, Eukaryotic
Centromere
Chromosome, Eukaryotic
Chromosome, Prokaryotic
Crossing Over
DNA Polymerases
DNA Repair
Embryonic Stem Cells
Eubacteria
Inheritance, Extranuclear
Linkage and Recombination
Meiosis
Mitosis
Oncogenes
Operon
Polyploidy
Replication
Signal Transduction
Telomere
Tumor Suppressor Genes

CLONED OR TRANSGENIC ORGANISMS

Agricultural Biotechnology
Biopesticides
Biotechnology
Biotechnology: Ethical Issues
Cloning Organisms
Cloning: Ethical Issues
Gene Targeting
Model Organisms
Patenting Genes
Reproductive Technology
Reproductive Technology: Ethical Issues
Rodent Models
Transgenic Animals
Transgenic Microorganisms
Transgenic Organisms: Ethical Issues
Transgenic Plants

DEVELOPMENT, LIFE CYCLE, AND NORMAL HUMAN VARIATION

Aging and Life Span
Behavior
Blood Type
Color Vision
Development, Genetic Control of
Eye Color
Fertilization
Genotype and Phenotype
Hormonal Regulation
Immune System Genetics
Individual Genetic Variation
Intelligence
Mosaicism
Sex Determination
Sexual Orientation
Twins
X Chromosome
Y Chromosome

DNA, GENE AND CHROMOSOME STRUCTURE

DNA TECHNOLOGY

ETHICAL, LEGAL, AND SOCIAL ISSUES

GENE DISCOVERY

GENE EXPRESSION AND REGULATION

GENETIC DISORDERS

GENETIC MEDICINE: DIAGNOSIS, TESTING, AND TREATMENT

GENOMES

GENOMICS, PROTEOMICS, AND BIOINFORMATICS

HISTORY

INHERITANCE

MODEL ORGANISMS

MUTATION

ORGANISMS, CELL TYPES, VIRUSES

POPULATION GENETICS AND EVOLUTION

RNA